

Week 20: Real-Time Features & File Uploads

1. WebSockets (Socket.io) Introduction

WebSockets enable real-time two-way communication between client and server. Socket.io is a library that simplifies WebSocket implementation in Node.js applications. It is commonly used in chat apps, notifications, and live systems.

2. Real-Time Chat / Notifications

Real-time chat and notification systems instantly update data without page refresh. This improves responsiveness and user interaction. These features are widely used in messaging and social media applications.

3. File Uploads (Multer)

Multer is middleware used in Node.js for handling file uploads. It allows users to upload images, documents, videos, and attachments to the server. This is useful for profile pictures, media sharing, and document storage.

4. Form Handling & Validation (React Hook Form)

React Hook Form is a library used for managing forms in React applications. It simplifies form validation, state management, and submission handling. It improves performance and reduces unnecessary re-renders.

5. Real-Time Data Synchronization

Real-time synchronization ensures that changes made by one user instantly reflect for others. This is useful in chat systems, collaborative tools, and live dashboards.

Conclusion

This week focused on advanced full-stack features. By learning WebSockets, Socket.io, real-time communication, file uploads, form validation, and synchronization, stronger backend and real-time application development skills were built.